Volume Cube Stack Task Cards

Unit 10: Time Capsule Engineers · Lesson 10-1 · EduWonderLab

UNIT 10 · LESSON 10-1 · TEACHER NOTES | **Standard 6.G.2**

FORMAT **Build + count task cards**	TIME **25–30 min**
GROUPING **Pairs**	TOPIC **Volume with Whole Number Edges**
STANDARD **6.G.2**	PREP LEVEL **Light — print prism cards and layer organizer**

Learning goal *I can find the volume of a rectangular prism by counting unit-cube layers.*

Teacher setup

- Print one card set and organizer per pair. Each card shows a prism's dimensions.
- Students count the cubes in one layer, then multiply by the number of layers.
- Connect counting to length × width × height.

Materials

Unit-cube prism cards, Layer organizer, Pencil.

Differentiation & language support

- Layer organizer: cubes per layer × number of layers (SPED).
- Sentence stem: "There are ___ layers with ___ cubes in each layer."
- ESOL: shade one layer on the prism to model the idea.

Key vocabulary (English / Español):

Term	Español	What it means
volume	volumen	the space inside a solid, in cubic units
unit cube	cubo unitario	a cube that is 1 by 1 by 1

Quick teacher check: *Ask a pair to show the cubes-per-layer step on Card 5.*

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Name: **Date:** **Class:**

Your goal: *I can find the volume of a rectangular prism by counting unit-cube layers.*

What you need

Prism cards, Layer organizer.

How to play

1. Pick a prism card and find the cubes in one layer.
2. Count the number of layers (the height).
3. Multiply to find the volume.
4. Record the volume in cubic units.

Prism Cards

Find the cubes in one layer (length × width), then multiply by the height.

1. length 4, width 3, height 2	**2.** length 5, width 2, height 3
3. length 6, width 4, height 2	**4.** length 3, width 3, height 3
5. length 5, width 5, height 2	**6.** length 7, width 2, height 4
7. length 8, width 3, height 2	**8.** length 10, width 2, height 3

Record your work (cubes per layer × layers):

Explain your thinking

Explain how counting layers is the same as length × width × height.

Sentence frame: *There are ___ layers with ___ cubes in each layer, so the volume is ___.*

★ **Challenge:** *Cards 3 and 7 both equal 48. Find a third prism with whole edges and volume 48.*

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. length 4, width 3, height 2 **Answer: 24 cubic units**	**2.** length 5, width 2, height 3 **Answer: 30 cubic units**
3. length 6, width 4, height 2 **Answer: 48 cubic units**	**4.** length 3, width 3, height 3 **Answer: 27 cubic units**
5. length 5, width 5, height 2 **Answer: 50 cubic units**	**6.** length 7, width 2, height 4 **Answer: 56 cubic units**
7. length 8, width 3, height 2 **Answer: 48 cubic units**	**8.** length 10, width 2, height 3 **Answer: 60 cubic units**

Teaching notes & look-fors

- Volume = length × width × height = (one layer) × (number of layers).
- Cards 3 and 7 both equal 48 cubic units.